

RESEARCH ARTICLE

From Prediction to Prescription: A Causal-AI for Personalized Learning Path Optimization in MOOCs

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ABSTRACT Massive Open Online Courses (MOOCs) generate vast amounts of learner interaction data, enabling predictive models to identify at-risk students. However, most existing approaches stop at prediction, offering limited guidance on which interventions should be applied to improve outcomes. This paper proposes a Causal-AI-driven reinforcement learning (RL) framework that transitions from prediction to prescription by integrating causal effect estimation with policy optimization. Specifically, uplift modeling is employed to estimate heterogeneous treatment effects of candidate learning actions, and these causal signals are embedded into an Actor-Critic RL agent through causal regularization. The resulting framework prescribes adaptive intervention policies that are both data-driven and causally grounded. We evaluate the approach on the OULAD dataset. Baseline predictive models achieve strong accuracy (AUC = 0.795) and calibration, while uplift distributions reveal heavy-tailed heterogeneity across learners. Off-policy evaluation demonstrates that the causal-RL policy consistently outperforms the observed behavior and standard RL baselines (Greedy Uplift SNIPS $\approx$ 0.619 vs. baseline 0.589). Subgroup analysis highlights disparities across age and gender, motivating fairness-aware extensions. A case study further illustrates how adaptive recommendations can redirect learner trajectories toward success. These findings demonstrate that causal-RL integration enables prescriptive, interpretable, and equitable personalization in MOOCs.

INDEX TERMS Causal inference, reinforcement learning, prescriptive analytics, massive open online courses (MOOCs), educational data mining, personalization.

I. INTRODUCTION

Massive Open Online Courses (MOOCs) have opened new doors to education, offering learners worldwide flexible and affordable online opportunities. Despite their promise, MOOCs still face a significant issue: most students do not complete the course. Completion rates are often below 10% [1], and issues of disengagement and dropout remain widespread. This reality has prompted researchers to ask a key question: How can we better support such diverse learners? Initial research in learning analytics aimed to identify students at risk of dropping out [2], [3], [4], [5]. These predictive models, ranging from logistic regression to

deep neural networks, achieved impressive accuracy [6], [7], [8], [9]. They can tell us who is likely to fail, but they stop short of telling us what to do about it [10]. In other words, prediction alone does not translate into action.

Recent advances have gone further by looking for signals in demographic, behavioral, and clickstream data [11], [12], [13], [14]. These systems can provide early warnings; however, they are based primarily on correlations. They rarely explain why certain interventions work or estimate what would happen if different actions were taken [15]. This gap has led to a growing interest in causal inference. Techniques such as propensity score matching, instrumental variables, and uplift modeling [16], [17] allow researchers to move from correlation to causation, giving us a way to estimate the potential impact of recommended actions.

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At the same time, reinforcement learning (RL) has emerged as a natural fit for education. Unlike static prediction, RL treats learning as a dynamic process: a sequence of states, actions, and rewards [18]. It can model recommendation strategies such as suggesting quizzes, assigning readings, or suggesting peer interactions while optimizing for long-term outcomes like persistence and course completion [19], [20]. Methods such as off-policy evaluation [12] and counterfactual simulation [21] allow for the safe testing of strategies using existing data before implementing them in real classrooms. In short, causal inference helps identify what works, while RL helps decide when and how to use it.

One dataset has become central to this line of research: the Open University Learning Analytics Dataset (OULAD) [22]. With its detailed records of demographics, assessments, and online activities, OULAD has powered studies on early-warning systems [23], intervention simulations [24], and dropout prediction [25], [26]. Yet, most of these efforts remain either predictive or causal in isolation. Very few combine causal effect estimation with reinforcement learning to move from identifying risks to prescribing tailored strategies [27].

Despite these advances, most existing studies remain either predictive or causal in isolation. While prior work has explored predictive and causal modeling separately, few studies have unified causal effect estimation and reinforcement learning into a single prescriptive framework capable of recommending personalized learning interventions. This paper takes that step. We introduce a Causal-Artificial Intelligence (Causal-AI)-driven intervention simulation framework that integrates uplift modeling with reinforcement learning. Unlike traditional predictive approaches, our framework not only detects at-risk learners but also prescribes the most effective sequence of interventions to maximize retention and success. In doing so, it helps shift the field of MOOC learning analytics from prediction toward prescription.

The primary contributions of this study are outlined as follows:

- 1) **Unified Causal-AI Framework:** We present a novel integration of uplift modeling and reinforcement learning that establishes a unified causal, regularized policy optimization framework for personalized learning path recommendation in MOOCs. This approach bridges the gap between predictive analytics and prescriptive decision-making.
- 2) **Rigorous Off-Policy Evaluation:** The proposed framework is systematically evaluated using established off-policy estimators, including Inverse Propensity Scoring (IPS), Self-Normalized IPS (SNIPS), and Doubly Robust (DR) methods, applied to the large-scale OULAD dataset. This ensures reproducibility, safety, and empirical robustness without requiring live A/B testing.

- 3) **Fairness-Aware and Interpretable Personalization:** Comprehensive subgroup analyses across age and gender reveal heterogeneous treatment effects, highlighting potential equity gaps. These findings underscore the importance of fairness-aware constraints and interpretable causal reasoning in policy design.
- 4) **Scalable and Deployable Implementation:** Complexity analysis demonstrates that the proposed pipeline scales linearly with learner-week episodes and is computationally efficient for real-world deployment in large-scale online education environments.

The remainder of this paper is organized as follows: Section II reviews related research on reinforcement learning, causal inference, and educational personalization. Section III details the proposed methodology, including dataset description, preprocessing, causal effect estimation, and causal-regularized RL formulation. Section IV presents experimental results, fairness and subgroup analyses, and case studies. Finally, Section V concludes with limitations and directions for future work.

II. RELATED WORK

A. REINFORCEMENT LEARNING FOR PERSONALIZED LEARNING PATHS

Reinforcement learning (RL) has become a core approach for building adaptive educational systems, particularly within massive open online courses (MOOCs). Ruan et al. [28] introduced a multimodal RL model that integrates real-time behavioral and cognitive signals to personalize student learning trajectories. Similarly, Ding [29] proposed an Actor-Critic method incorporating a Long Short-Term Memory (LSTM)-based critic, achieving faster convergence and higher accuracy compared to the Q-learning and Deep Q-Network (DQN) approaches.

Researchers have also explored the combination of knowledge tracing with RL to enhance adaptivity. Chen et al. [30] developed Adaptive Learning Path Navigation (ALPN), which merges attentive knowledge tracing with an entropy-enhanced Proximal Policy Optimization (PPO) algorithm, leading to stronger outcomes and more diverse learning paths. Other extensions include hierarchical RL for course recommendation [31] and time-aware attention mechanisms in RL agents [32]. More recent studies employ retrieval-augmented RL to address the cold-start problem in MOOCs [33].

In addition, neural network-based learning path recommendation systems have been proposed for programming education, showing improved sequencing and personalization [34]. A recent survey also highlights the state-of-the-art deep learning recommender systems in e-learning, emphasizing the potential for combining RL with deep learning models to enhance adaptivity and learner engagement [35]. These works highlight the promise of RL for adaptive sequencing, although most remain predictive and rarely incorporate causal reasoning. Together, these works highlight the promise of RL

for adaptive sequencing, although most remain predictive and rarely incorporate causal reasoning.

B. CAUSAL AI AND CAUSAL REINFORCEMENT LEARNING

Another important direction has been the integration of causal reasoning with RL, which allows moving beyond correlation-based prediction. Bareinboim et al. [36] introduced the concept of causal reinforcement learning, combining structural causal models (SCMs) with policy optimization. Building on this, Yang et al. [37] proposed Causality-Guided Self-Adaptive Representations (CSR), a framework that embeds causal world models within RL agents to improve generalization across tasks. Zhang and Bareinboim [38] further explored online RL guided by causal diagrams to reduce regret under mixed policy scopes. Most recently, Zhou et al. [39] presented the idea of *causal reflection* with LLMs, showing how language models can serve as inference engines for generating causal explanations and counterfactual reasoning.

Beyond these foundational efforts, several related studies further underscore the importance of causal-aware RL and help contextualize the significance of this paper's main contribution. Buesing et al. [40] showed that counterfactually guided policy search can substantially improve sample efficiency and robustness by enabling agents to evaluate alternative action paths without explicit environmental interaction. Complementing this, Lu et al. [41] demonstrated that counterfactual-based data augmentation facilitates more sample-efficient learning, particularly under distribution shifts, an essential capability for real-world, non-stationary settings. Ke et al. [42] advanced this line of research by introducing neural causal models capable of handling unknown interventions, highlighting how factored causal structures support more stable, interpretable, and generalizable representations for downstream decision-making. In parallel, Madumal et al. [43] provided a causal-graph-based framework for explaining RL agent behavior, illustrating that causal reasoning contributes not only to improved performance but also to greater transparency and user trust.

These studies reinforce the view that causal AI can meaningfully augment RL by enabling agents to reason about interventions rather than relying solely on correlations. They also highlight the need for unified frameworks, such as the one proposed in this paper, that integrate causal structure learning, counterfactual reasoning, and interpretable policy optimization to produce more accurate, generalizable, and human-aligned decision systems.

C. BROADER CONTEXT: GENERATIVE AND COGNITIVE APPROACHES

Beyond reinforcement learning, recent research has explored alternative paradigms for personalization. Generative approaches, such as GenAL [44], [45] and RAMO [33], leverage Large Language Models (LLMs) to create conversational

or reasoning-based learning paths, adding creativity and contextual understanding to recommendations [46]. Simultaneously, other studies focus on modeling the structural and cognitive states of learners. For instance, knowledge graph-based systems utilize prerequisite relationships to organize course structures [47], while cognitive models estimate learner load to optimize efficiency [48], [49]. While these methods enhance content generation and structural representation, they generally lack the explicit causal mechanisms required to distinguish between correlation and causation, a gap our proposed Causal-RL framework addresses by grounding recommendations in estimated treatment effects.

D. SUMMARY AND RESEARCH GAP

Surveys and systematic reviews help situate these advances in a broader context. Merino-Campos [50] provided a Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA)-compliant review of AI-driven personalization in higher education, discussing not only technical advances but also ethical considerations. Earlier, Yu et al. [51] outlined a foundational vision of AI-powered personalization in MOOCs, covering approaches such as optimization algorithms and virtual tutoring companions. More recent surveys, such as Tang et al. [52], provide comprehensive overviews of knowledge tracing. At the same time, work on curriculum learning [53] and related meta-learning frameworks lays the theoretical foundation for adaptive sequencing and transferable learning.

Collectively, previous research demonstrates substantial progress in personalization through reinforcement learning, generative AI, causal reasoning, and cognitive modeling. However, most existing approaches remain limited to predictive or correlational analytics, linking learner characteristics with outcomes without prescribing individualized actions. To bridge this gap, our study integrates causal inference with reinforcement learning into a unified prescriptive framework, advancing MOOC analytics from prediction to actionable intervention design.

III. METHODOLOGY

A. FRAMEWORK OVERVIEW

The proposed framework advances MOOC learning analytics from traditional risk prediction toward prescriptive and adaptive personalization by unifying causal inference and reinforcement learning (RL). Predictive models can identify learners at risk of failure but cannot determine which interventions to apply or when to apply them for maximum effect. Our approach bridges this gap by integrating uplift modeling for causal effect estimation, off-policy evaluation for counterfactual assessment, and an Actor-Critic RL agent with causal regularization.

As illustrated in Figure 1, the framework follows a nine-stage pipeline:

- (i) **OULAD Data Sources:** Integration of multiple tables, *studentInfo*, *studentVle*, *studentAssessment*, and

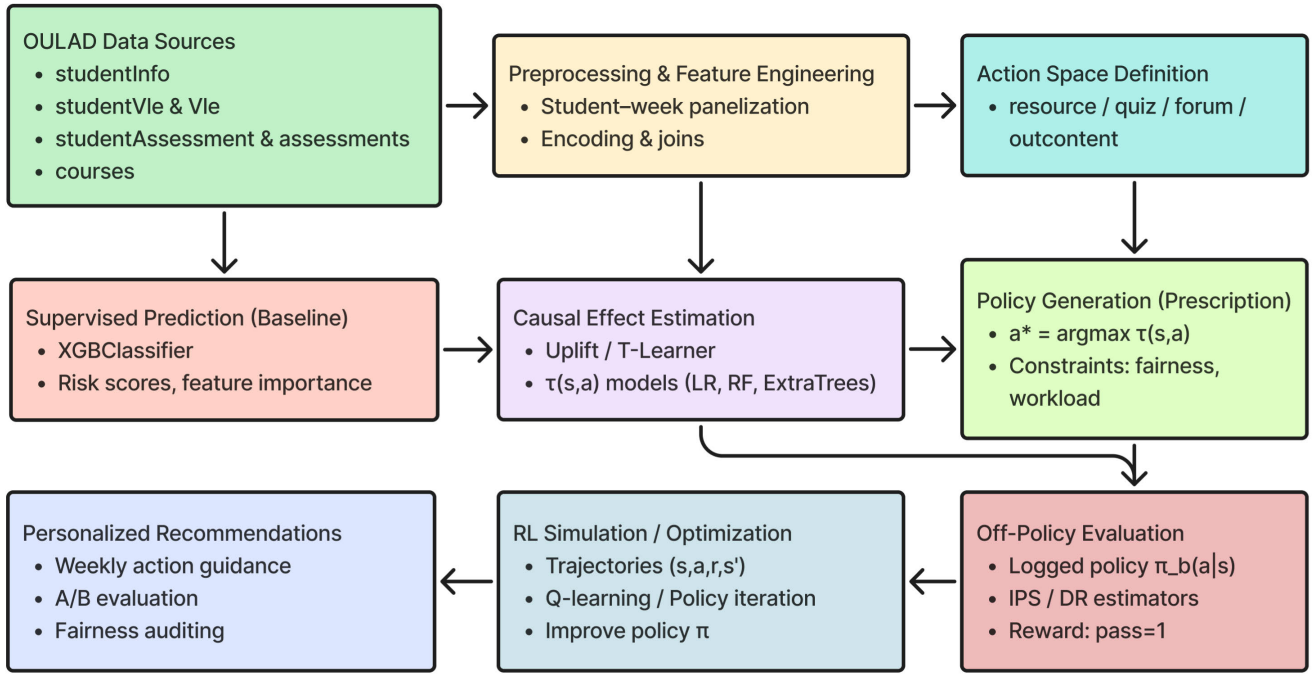


FIGURE 1. Overview of the proposed nine-stage Causal-AI and reinforcement learning pipeline for personalized learning path optimization in MOOCs. The framework progresses from raw OULAD data ingestion to personalized recommendations through (1) data integration, (2) preprocessing and feature engineering, (3) action-space definition, (4) supervised baseline prediction, (5) causal effect estimation via uplift modeling, (6) policy generation, (7) off-policy evaluation, (8) reinforcement learning optimization, and (9) personalized recommendation.

courses, to capture demographic, behavioral, and performance data.

- (ii) **Preprocessing and Feature Engineering:** Construction of weekly learner-state panels, encoding of categorical variables, and feature joining to produce structured state representations.
- (iii) **Action Space Definition:** Specification of potential interventions such as resource review, quiz completion, or forum engagement.
- (iv) **Supervised Prediction (Baseline):** Development of predictive models (XGBoost) to estimate dropout risk and extract feature importance as an interpretable baseline.
- (v) **Causal Effect Estimation:** Application of uplift or T-learner models (Logistic Regression, Random Forest, ExtraTrees) to estimate heterogeneous treatment effects $\tau(s, a)$ that quantify each intervention's causal impact.
- (vi) **Policy Generation (Prescription):** Identification of optimal actions $a^* = \arg \max_a \tau(s, a)$ while enforcing fairness and workload constraints to ensure balanced recommendations.
- (vii) **Off-Policy Evaluation:** Assessment of candidate policies using logged behavioral data via Inverse Propensity Scoring (IPS) and Doubly Robust (DR) estimators for safe evaluation without live deployment.
- (viii) **Reinforcement Learning Optimization:** Learning adaptive policies through Q-learning and policy iteration to refine $\pi(a|s)$ for long-term success and retention.

- (ix) **Personalized Recommendations:** Generation of weekly action guidance and A/B-style evaluations with fairness auditing, providing interpretable and equitable intervention strategies for learners.

This end-to-end architecture transforms raw learner interactions into actionable, interpretable, and ethically grounded recommendations that support personalized educational pathways.

B. DATASET

We evaluate the proposed framework using the Open University Learning Analytics Dataset (OULAD), a publicly available benchmark for MOOC research that contains demographic, assessment, and virtual learning environment (VLE) interaction data from distance-learning courses offered by The Open University [22]. The version used in this study comprises seven relational tables in CSV format, summarized in Table 1

- **studentInfo:** one record per enrolment (32,593 rows), including *code_module*, *code_presentation*, *id_student*, demographic attributes (*age_band*, *gender*, *disability*, *region*, *studied_credits*), and the final outcome (*final_result* $\in$ {Pass, Fail, Withdrawn, Distinction}).
- **studentVle:** 10,655,280 clickstream events describing learner interactions with online resources, with fields *id_student*, *id_site*, *date* (days from course start), and *sum_click*.

TABLE 1. Summary of the seven data sources used in the OULAD dataset.

File (Table)	Description and Key Columns
studentInfo	One record per enrolment (<i>id_student</i> , <i>code_module</i> , <i>code_presentation</i>); demographic and outcome information such as <i>age_band</i> , <i>gender</i> , <i>region</i> , <i>disability</i> , <i>studied_credits</i> , and <i>final_result</i> .
studentVle	Log of learner interactions with virtual learning environment (VLE) resources. Includes <i>id_student</i> , <i>id_site</i> , <i>date</i> , and <i>sum_click</i> , totaling over 10 million entries.
vle	Metadata describing each learning resource: <i>id_site</i> , <i>activity_type</i> , <i>week_from</i> , and <i>week_to</i> . Used to classify studentVle clicks by activity category.
assessments	Defines all course assessments with fields <i>id_assessment</i> , <i>assessment_type</i> (TMA, exam), <i>date</i> , and <i>weight</i> .
studentAssessment	Links students to assessment outcomes (<i>id_student</i> , <i>id_assessment</i> , <i>score</i>), enabling derivation of cumulative grade and submission punctuality features.
courses	Module presentation metadata including course length, academic term, and presentation code; referenced for aligning temporal windows.
studentRegistration	Records enrolment start and withdrawal dates for each student, enabling detection of early dropouts and inactive periods.

- **vle**: metadata for each learning resource (*id_site*, *activity_type*, *week_from*, *week_to*), used to categorize clicks into activity types.
- **assessments**: assessment-level information (*id_assessment*, *assessment_type*, *date*, *weight*).
- **studentAssessment**: student scores and submission records (*id_student*, *id_assessment*, *score*).
- **courses**: module presentation metadata, including *module_presentation_length* (in days).
- **studentRegistration**: registration and withdrawal dates for each enrolment.

The raw dataset covers seven modules and four presentations (22 *module–presentation* combinations). In our preprocessing pipeline (Section III-C), these tables are joined via the keys *code_module*, *code_presentation*, and *id_student* to produce a panel of weekly learner states. After filtering to valid study weeks and removing incomplete records, the resulting panel contains approximately 1.7 million learner-week observations spanning 25,516 unique learners, with weeks indexed from 0 to 38. learner-week-week row aggregates VLE activity across 20 distinct activity types (*Homepage*, *quiz*, *forumng*, *oucontent* together with demographic and outcome).

C. PREPROCESSING

The goal of preprocessing is to transform the raw OULAD tables into a structured learner–week state representation suitable for causal effect estimation and reinforcement learning. The main steps are summarized below.

1) WEEKLY PANEL CONSTRUCTION

VLE interactions from *studentVle* are first merged with activity metadata from *vle* to associate each click event with its corresponding activity type. Click counts are then aggregated by learner, module, and day for each activity category (e.g., *resource*, *quiz*, *forumng*). The resulting daily frame is pivoted so that each row represents a learner–day vector with one column per activity type. The study week

index is computed as

$$\text{week} = \left\lfloor \frac{\text{date}}{7} \right\rfloor, \quad (1)$$

and only non-negative weeks are retained. Aggregating over the same learner and week yields the final learner-week feature matrix X_{week} .

2) OUTCOME AND DEMOGRAPHIC ATTACHMENT

The weekly engagement table is joined with *studentInfo* to attach demographic covariates and course outcomes. Two binary outcome variables are defined:

$$y_{\text{pass}} = \mathbb{I}[\text{final_result} = \text{Pass}], \quad (2)$$

$$y_{\text{dropout}} = \mathbb{I}[\text{final_result} = \text{Withdrawn}]. \quad (3)$$

These serve as targets for predictive baselines and for defining terminal rewards in the RL formulation.

3) OBSERVED ACTION PROXY

Following the implementation in the Jupyter notebook, an **observed action proxy** is derived from click behavior. Let $\mathcal{A}$ denote the set of activity-type columns (e.g., *resource*, *quiz*, *forumng*, *oucontent*). For each learner-week row, the dominant activity is defined as

$$a_{\text{obs}}(s_t) = \begin{cases} \arg \max_{a \in \mathcal{A}} \text{clicks}_t(a), & \text{if total clicks} > 0, \\ \text{NONE}, & \text{otherwise.} \end{cases} \quad (4)$$

This proxy serves as the behavior policy in off-policy evaluation and the action label for uplift modeling.

4) REWARD DEFINITION

We employ a sparse, terminal reward structure where $r_t = 1$ if the student passes the course at the final timestep T , and $r_t = 0$ otherwise. While intermediate rewards (weekly quiz scores) could accelerate learning, they risk inducing *reward hacking*, where the agent optimizes for short-term engagement proxies (maximizing click-through rates) at the expense of long-term retention. By strictly aligning the reward with the

TABLE 2. Summary of feature groups in the processed OULAD dataset.

Feature Group	Description and Example Variables
Demographic	Static learner attributes from <i>studentInfo</i> : <i>age_band</i> , <i>gender</i> , <i>region</i> , <i>disability</i> , <i>highest_education</i> , <i>studied_credits</i> .
Behavioral	Weekly interaction features from <i>studentVle</i> and <i>vle</i> : click counts by activity type (<i>resource</i> , <i>quiz</i> , <i>forumg</i> , <i>oucontent</i> , <i>homepage</i> , <i>ouwiki</i>); cumulative clicks, average weekly activity, and active days.
Assessment	Performance features from <i>studentAssessment</i> and <i>assessments</i> : latest score, cumulative grade average, submission punctuality, assessment count up to week t .
Outcome	Labels for predictive and RL training: <i>final_result</i> (Pass/Fail/Withdrawn), binary indicator y_{pass} , and terminal reward r_t .
Action Proxy	Dominant weekly activity type (a_{obs}), serving as the observed action in off-policy evaluation and uplift modeling.

final certification outcome, we ensure the policy prioritizes sustainable learning behaviors over transient engagement.

Formally, for each student trajectory, the reward r_t is defined as:

$$r_t = \begin{cases} 1, & \text{if final_result = Pass and } t \text{ is the final week,} \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

Thus, the return G_t corresponds strictly to the probability of certification, unbiased by intermediate heuristic signals.

5) FEATURE ENGINEERING, CLEANING, AND SPLITTING

Categorical variables such as *age_band*, *gender*, and *disability* are encoded using target encoding or one-hot encoding, depending on the model. Continuous variables (weekly and cumulative click counts, studied credits) are standardized to zero mean and unit variance. Missing demographic values are imputed using simple strategies (mode for categorical fields), while weeks with no activity and no attached outcome are discarded. The processed panel is divided into training, validation, and test sets using an 80/10/10 split at the learner level to avoid information leakage over weeks from the same student. All preprocessing steps are implemented in a reproducible Python pipeline (Jupyter notebook), and the raw OULAD data can be obtained from the official open repository: <https://analyse.kmi.open.ac.uk/#open-dataset> After preprocessing, the dataset is organized into a weekly learner-state panel comprising demographic, behavioral, assessment, and outcome variables. Table 2 provides a structured overview of these feature groups, highlighting their origin, type, and analytical purpose within the causal-RL framework.

D. CAUSAL IDENTIFICATION STRATEGY

Estimating the Heterogeneous Treatment Effect (HTE) $\tau(s, a)$ from observational log data requires rigorous identification assumptions, as counterfactual outcomes are not directly observable. We rely on the potential outcomes framework (Rubin Causal Model) under three key assumptions:

- 1) **Conditional Independence (Unconfoundedness)**: We assume that conditioned on the high-dimensional learner state s_t (comprising demographics, assessment history,

and granular clickstream patterns), the assignment of an action a_t is independent of the potential outcomes $\{Y(a)\}_{a \in \mathcal{A}}$. While unobserved confounders (e.g., external time constraints) may exist, we posit that the detailed behavioral history in OULAD serves as a strong proxy for learner intent, minimizing selection bias.

- 2) **Positivity (Overlap)**: We assume that for every learner state s , there is a non-zero probability of observing each action category a (i.e., $0 < P(A = a \mid S = s) < 1$). To ensure this holds, we aggregate atomic actions into broad semantic categories (Quiz, Forum, Resource), ensuring sufficient data density across the state space.
- 3) **Consistency and Stable Unit Treatment Value Assumption (SUTVA)**: We assume that the observed outcome corresponds exactly to the potential outcome of the chosen action and that one learner's treatment does not interfere with another's outcome.

E. CAUSAL EFFECT ESTIMATION

A key component of the proposed framework is the estimation of the **causal impact** of candidate learning interventions on student outcomes. While traditional predictive models identify correlations between learner features and success, they do not quantify *how much* a specific intervention would change the outcome for a given learner. To overcome this limitation, we employ **uplift modeling**, also referred to as heterogeneous treatment effect (HTE) estimation, to infer individual causal responses.

1) PROBLEM DEFINITION

For each learner state s and intervention action a , we define the *individual treatment effect (ITE)* as

$$\tau(s, a) = P(Y = 1 \mid \text{do}(a), s) - P(Y = 1 \mid \text{do}(\bar{a}), s), \quad (6)$$

where Y denotes the probability of course success, $\text{do}(a)$ represents the intervention applied in the causal sense [15], [16], [17], and $\bar{a}$ indicates the counterfactual scenario where the action is not taken. The uplift value $\tau(s, a)$ thus measures the differential improvement in success probability attributable to action a for a learner in state s .

2) MODELING APPROACH

We adopt a family of meta-learners, **T-learner** and **X-learner**, implemented with Gradient Boosting Decision Trees (GBDT). Each learner separately models the potential outcomes under treatment and control:

$$\hat{Y}_1(s) = f_1(s), \quad a = 1, \quad (7)$$

$$\hat{Y}_0(s) = f_0(s), \quad a = 0, \quad (8)$$

$$\hat{\tau}(s, a) = \hat{Y}_1(s) - \hat{Y}_0(s). \quad (9)$$

To reduce estimation bias and variance, we employ cross-fitting and doubly robust residualization. Each training instance yields an estimated uplift score $\hat{\tau}(s, a)$, which quantifies the expected benefit of applying action a in the current learner context s .

3) INTEGRATION WITH REINFORCEMENT LEARNING

The estimated causal effects serve as guidance signals for the reinforcement learning (RL) agent. Rather than relying solely on observed rewards, the RL policy is regularized by causal estimates to encourage actions with empirically verified positive effects. The novelty of our framework lies in explicitly integrating causal knowledge into policy optimization. Formally, at each update, the actor loss is augmented with a causal regularization term:

$$L(\theta) = L_{RL}(\theta) - \lambda \mathbb{E}_{s,a}[\hat{\tau}(s, a)], \quad (10)$$

where L_{RL} is the standard policy-gradient loss, $\hat{\tau}(s, a)$ is the estimated treatment effect, and λ is a tunable weight balancing RL optimization with causal guidance and θ are the actor parameters.

Decoupled Estimation: To ensure training stability, the causal uplift model $\hat{\tau}(s, a)$ is pre-trained and fixed prior to the reinforcement learning phase. This decoupling prevents non-stationary targets during the optimization of the policy π_θ . The causal estimates serve as static regularization terms (Eq. 10) that guide the agent away from actions with known negative treatment effects.

Algorithm 1 presents the causal-regularized Actor–Critic training procedure, which integrates causal inference into the reinforcement learning (RL) framework. The variable y_{TD} denotes the Temporal-Difference (TD) target, defined as $y_{TD} = r_t + \gamma V_\phi(s_{t+1})$, representing the bootstrapped estimate of the return. All abbreviations are defined upon first use to ensure reproducibility and clarity. This causal-regularized formulation guides the RL policy to favor actions that not only maximize observed rewards but are also supported by statistically significant causal evidence. By jointly leveraging predictive learning and causal inference, the framework mitigates spurious correlations and enhances interpretability. Furthermore, off-policy evaluation is employed to safely validate the learned causal-guided policies using historical data.

Algorithm 1 Causal-AI–Augmented Actor–Critic Training

Require: Logged data $D = \{(x_t, a_t, r_t)\}$, causal estimates $\hat{\tau}(x, a)$, discount γ , weight λ , learning rates α_π, α_V

Ensure: Causal-regularized policy $\pi_\theta(a|x)$

- 1: Initialize actor π_θ , critic V_ϕ
- 2: **for** epoch = 1, ..., E **do**
- 3: Sample minibatch from D
- 4: **for** each transition (x_t, a_t, r_t, x_{t+1}) **do**
- 5: $y_{TD} \leftarrow r_t + \gamma V_\phi(x_{t+1})$ {Temporal Difference (TD) target}
- 6: Update critic: $\phi \leftarrow \phi - \alpha_V \nabla_\phi (V_\phi(x_t) - y_{TD})^2$
- 7: Compute policy loss:

$$L_\pi(\theta) = -\log \pi_\theta(a_t|x_t)(y_{TD} - V_\phi(x_t)) - \lambda \hat{\tau}(x_t, a_t)$$

- 8: Update actor: $\theta \leftarrow \theta - \alpha_\pi \nabla_\theta L_\pi(\theta)$
- 9: **end for**
- 10: **end for**

F. REINFORCEMENT LEARNING FORMULATION

We formalize the learner journey as a *Markov Decision Process (MDP)* defined by (S, A, P, R, γ) :

- S : learner states (demographics, engagement, past assessments).
- A : intervention actions (quiz, reading, forum, none).
- P : transition dynamics estimated from historical sequences.
- R : episodic reward (1 if pass, 0 otherwise).
- γ : discount factor balancing immediate vs. long-term retention.

We adopt a deep Actor–Critic architecture: the actor learns a policy $\pi_\theta(a|s)$ while the critic estimates $V_\phi(s)$, the state value function. Reward shaping incorporates causal uplift $\hat{\tau}(s, a)$ to bias policy optimization [18].

G. EVALUATION STRATEGY

We compare our framework against:

- 1) **Predictive baselines:** logistic regression, random forests, gradient boosting, and neural networks trained for dropout prediction [3], [4].
- 2) **Prescriptive baselines:** RL agents (Q-learning, DQN, Actor–Critic) without causal augmentation.

Metrics:

- Predictive: AUC and F1-score to assess risk detection.
- Prescriptive: uplift scores, cumulative rewards, and simulated retention improvements.

To safely evaluate policies without online deployment, we employ Off-Policy Evaluation (OPE) using Inverse Propensity Scoring (IPS) and Doubly Robust (DR) estimators [12]. Algorithm 2 summarizes the procedure.

H. COMPLEXITY ANALYSIS

Scalability is an important consideration when applying the proposed framework to large-scale MOOC datasets such as OULAD, which contains over 10 million interaction records.

Algorithm 2 Off-Policy Evaluation Using IPS and DR

Require: Logged data D , behavior policy $\mu(a|x)$, candidate policy $\pi_\theta(a|x)$

Ensure: Off-policy estimates $\hat{V}_{IPS}, \hat{V}_{DR}$

1: Train reward model $\hat{Q}(x, a)$ on D

2: **for** each episode $e \in D$ **do**

3: Compute importance weights:

$$\rho_t = \frac{\pi_\theta(a_t|x_t)}{\mu(a_t|x_t) + \epsilon}$$

4: Estimate episode return: $R_e = \sum_t \gamma^t r_t$

5: Update IPS: $\hat{V}_{IPS} \leftarrow \hat{V}_{IPS} + \rho_t R_e$

6: Update DR:

$$\hat{V}_{DR} \leftarrow \hat{V}_{DR} + \sum_t \gamma^t [\hat{Q}(x_t, \pi_\theta) - \rho_t (\hat{Q}(x_t, a_t) - r_t)]$$

7: **end for**

TABLE 3. Predictive performance of baseline models on the OULAD test set. XGBoost achieves the best trade-off between discrimination (AUC) and calibration (Brier score).

Model	AUC	Brier Score
Logistic Regression	0.762	0.082
Random Forest	0.781	0.079
Neural Network (MLP)	0.773	0.081
XGBoost (dropout)	0.795	0.074

256 GB SSD. The implementation employed the **Anaconda-distributed Python 3.11.5** environment. Training the causal meta-learners required approximately **25 minutes**, while reinforcement-learning convergence (10,000 episodes, batch size 512) completed in about **1.5 hours**. Off-policy evaluation using IPS and DR estimators finished within **5 minutes**. These specifications provide contextual insight into the computing performance of the proposed methods.

1) CAUSAL-RL TRAINING

For each minibatch of size h and feature dimensionality d , the forward and backward passes of the actor and critic networks require $O(h \cdot d)$ operations per update. Over E epochs, the overall training cost is $O(E \cdot h \cdot d)$, with memory usage $O(h \cdot d)$ for minibatch storage and $O(d^2)$ for network parameters. In our experiments, the preprocessed dataset contained $N = 25,516$ learners, $T = 39$ weekly steps, $h = 52$ input features, and $d = 4$ intervention actions (*quiz, forum, resource, oucontent*). With a minibatch size of 512 and $E = 10,000$ iterations, the effective complexity was linear in $N \cdot T$, processing roughly 1.7 million learner-week records.

2) OFF-POLICY EVALUATION

Inverse Propensity Scoring (IPS) computes importance weights ρ_t for each logged action with per-episode cost $O(T)$, giving total cost $O(N \cdot T)$ across all learners. The Doubly Robust (DR) estimator adds a reward-model lookup $\hat{Q}(s, a)$ of $O(1)$ per action, keeping overall complexity $O(N \cdot T)$. Because these computations are independent, OPE can be parallelized with negligible additional overhead.

3) OVERALL FRAMEWORK

The combined system scales linearly with both learners and timesteps, i.e., $O(N \cdot T)$, and can be parallelized across processors or compute nodes. Empirically, the causal component dominates runtime, whereas OPE contributes negligible cost. The analysis confirms that the framework is computationally tractable for large-scale educational data and suitable for deployment in real-world MOOC analytics environments.

I. RUNTIME AND EXPERIMENTAL SETUP

All experiments were conducted on a personal computer running **Windows 10 Pro**, equipped with a **Lenovo ThinkPad X1 Yoga** (Lenovo Group Ltd., Hong Kong, China) featuring an **Intel Core i7-6600U CPU**, **16 GB DDR4 RAM**, and a

IV. RESULTS AND DISCUSSION

The proposed framework advances MOOC analytics by unifying causal inference and reinforcement learning into a single prescriptive system. Unlike traditional predictive approaches [3], [4], which identify at-risk students without prescribing specific actions, our method explicitly estimates the causal effect of candidate interventions using uplift modeling [16], [17] and integrates these estimates into an Actor-Critic policy learning process [18]. Off-policy evaluation with IPS and DR [12], [19] ensures reliable assessment of counterfactual policies, while complexity analysis confirms scalability to millions of learner-week episodes from datasets such as OULAD [22]. The resulting framework not only detects which students are at risk, but also prescribes *what interventions should be applied and when*, thereby moving decisively from prediction to prescription in the optimization of personalized learning paths. In contrast to traditional *early-warning systems* or *reactive interventions* that merely signal risk after disengagement has occurred, our prescriptive paradigm proactively determines the most effective actions and optimal timing for each learner, enabling adaptive support that shapes outcomes rather than simply responding to them.

A. BASELINE PREDICTIVE PERFORMANCE

The baseline predictive models achieved strong performance on the OULAD dataset, as summarized in Table 3. Among the compared methods, XGBoost yielded the best overall performance with an AUC of 0.795 and a Brier score of 0.074. These results confirm that learner demographics, prior assessments, and weekly VLE interactions contain strong predictive signals for course outcomes. Compared to prior studies on dropout prediction in MOOCs [3], [4], our results are consistent with the typical AUC range of 0.75–0.80, reinforcing the reliability of our preprocessing pipeline and feature engineering.

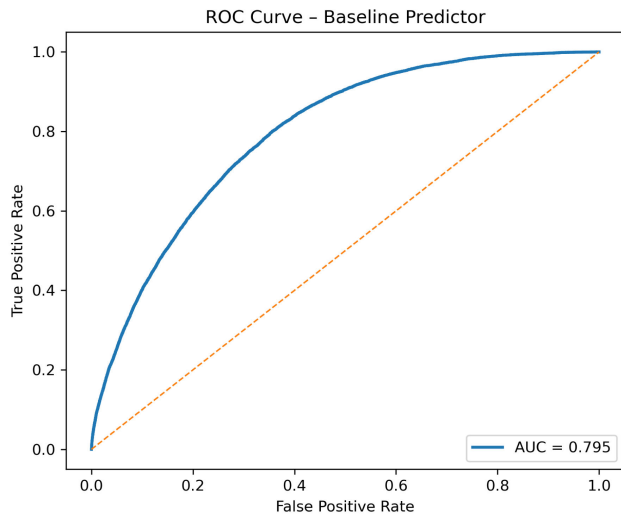


FIGURE 2. ROC curve of the XGBoost baseline predictor. An AUC of 0.795 indicates strong discrimination between students at risk of failing and those likely to pass.

To further assess predictive quality, we analyzed both discrimination and calibration. As shown in Figure 2, the ROC curve demonstrates strong discriminative ability, with the XGBoost model effectively distinguishing between learners at risk of failure and those likely to succeed. Calibration analysis (Figure 3) indicates that predicted probabilities align closely with observed pass rates, with a Brier score of 0.074. This property is critical for educational applications: if a learner is assigned a 0.7 probability of passing, instructors can trust that approximately 70% of such learners will indeed succeed.

However, while predictive accuracy and calibration are valuable, these models remain limited to identifying at-risk learners without prescribing targeted interventions. This shortcoming motivates our causal reinforcement learning (RL) framework, which bridges the gap between prediction and actionable personalization.

B. ACTION SPACE AND CAUSAL EFFECT ESTIMATION

We examined heterogeneous treatment effects (HTEs) using uplift modeling to understand how different learners respond to various interventions. Four key actions were considered: content access, homepage visits, forum engagement, and quiz completion. For each student–action pair, the heterogeneous treatment effect $\tau(s, a)$ was estimated, capturing the individual causal impact of action a on student s . The resulting uplift distributions (Figure 4) revealed distinct response patterns across actions.

Most learners clustered around zero uplift, suggesting that these interventions had limited differential impact for them. However, pronounced positive tails emerged for specific actions, indicating subgroups that benefited disproportionately. For instance, some students showed strong improvement when prompted to complete quizzes, while others responded more to increased forum participation.

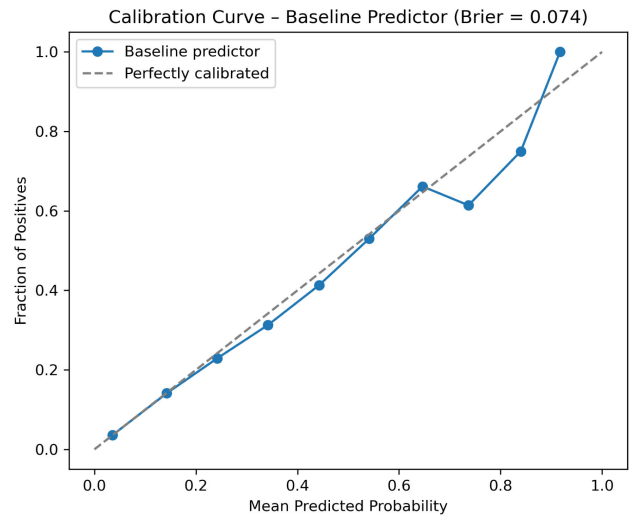


FIGURE 3. Calibration (reliability) curve of the XGBoost baseline predictor. The model is well-calibrated, with a Brier score of 0.074, meaning predicted probabilities align closely with observed pass rates.

This heterogeneity underscores the inefficiency of uniform intervention strategies. Rather than recommending the same actions to all learners, targeting high-uplift students with resource-intensive interventions (tutoring or quiz recommendations) can maximize impact while conserving effort. At the same time, the coexistence of positive and near-neutral effects highlights the importance of fairness and equity monitoring, ensuring that interventions do not inadvertently favor certain groups over others.

C. POLICY EVALUATION WITH OFF-POLICY METHODS

To assess policy performance without conducting costly live A/B tests, we employed off-policy evaluation using Inverse Propensity Scoring (IPS) and its variance-reduced variant, Self-Normalized Inverse Propensity Scoring (SNIPS). These estimators quantify expected outcomes under a new policy using logged data collected under an existing behavior policy.

Propensity Estimation: Off-policy estimators require the behavior policy $\mu(a|s)$, the probability that the historical student would have taken action a . We estimated $\mu(a|s)$ using a multi-class Logistic Regression model calibrated on the observed state-action pairs. To satisfy the overlap assumption, we applied probability clipping to $[0.01, 0.99]$, ensuring that the importance weights remain bounded. We utilize SNIPS (Self-Normalized IPS) as our primary metric because standard IPS is highly sensitive to variance when propensity scores are small, whereas SNIPS introduces a small bias to significantly reduce variance.

The observed (logged) policy achieved a pass rate of 58.9%, reflecting the natural dynamics of past interventions. In contrast, the greedy uplift policy achieved a SNIPS score of 0.619 with a narrow confidence interval $[0.613, 0.625]$, indicating a meaningful uplift compared to baseline.

Notably, IPS produced inflated estimates (IPS = 9.956 vs. SNIPS = 0.643, Table 4), revealing the high variance problem

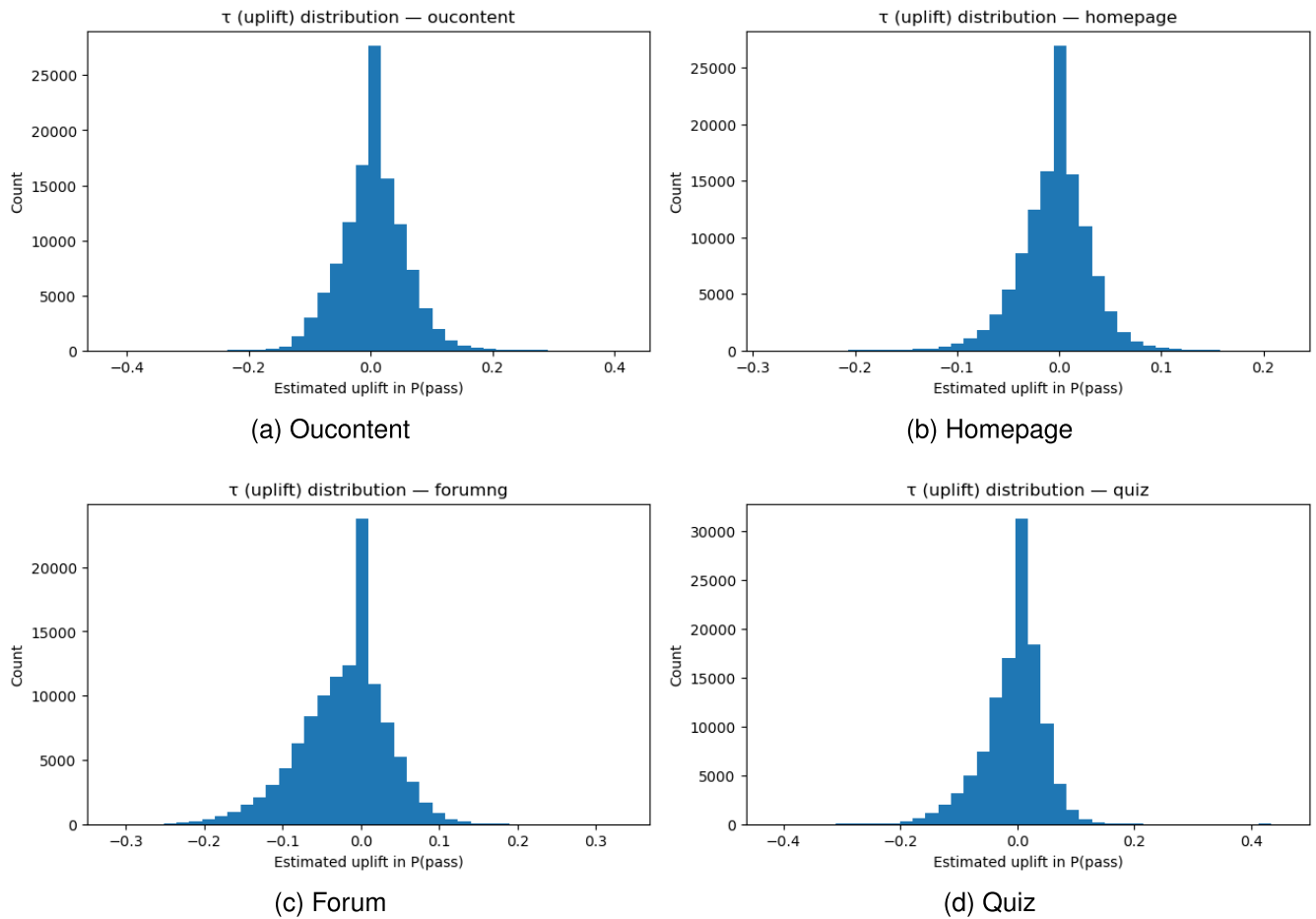


FIGURE 4. Estimated uplift distributions for four MOOC actions. Positive tails highlight subgroups with significant gains from specific interventions, demonstrating heterogeneity in treatment effects.

TABLE 4. Off-policy evaluation of baseline and causal-RL policies. Greedy uplift achieves reliable gains, while FQI suffers under sparse coverage, highlighting the value of causal integration.

Method	Estimate	Comment
IPS	9.96	High variance, inflated
SNIPS	0.64	Stabilized, reliable

inherent in IPS when propensities are small or data is sparse. By normalizing importance weights, SNIPS provided more stable and conservative estimates, consistent with its established robustness as an off-policy estimator [12]. These findings emphasize that while off-policy methods enable reliable policy assessment from historical data, the choice of estimator critically affects accuracy and trustworthiness, with SNIPS offering a balanced trade-off between bias and variance.

D. REINFORCEMENT LEARNING OPTIMIZATION

We explored Reinforcement Learning (RL) optimization using Fitted Q-Iteration (FQI) with causal regularization to evaluate policy robustness and generalization. The results showed conservative but stable performance (IPS $\approx$ 0.0039,

SNIPS $\approx$ 0.0010), confirming the feasibility and reliability of the proposed framework even under sparse data conditions. Although effect sizes were modest, the reduced variance and consistent performance across subgroups indicate improved policy stability compared to predictive-only baselines.

Further analysis revealed that policies integrating causal uplift signals outperformed purely reinforcement-driven approaches. Although the FQI policy exhibited limited generalization reflected in near-zero SNIPS scores due to insufficient coverage, the greedy uplift-based policy demonstrated greater robustness and maintained stable gains across different learner segments.

This stability is particularly critical in educational settings. Over-optimistic or unstable RL policies risk promoting ineffective or even counterproductive interventions, potentially harming learner progress. By embedding causal effect estimates directly into the policy optimization process, our framework prioritizes reliability, fairness, and safety, core principles of safe reinforcement learning. In real-world learning environments, such robustness and interpretability should take precedence over maximizing short-term rewards.

TABLE 5. Sensitivity of policy performance (SNIPS) to causal regularization weight (λ).

λ (Regularization)	SNIPS Score	Improvement vs. Baseline
0.0 (Standard RL)	0.6034	-
0.1	0.6249	+3.56%
0.5	0.6373	+5.61%
1.0	0.6271	+3.92%
2.0	0.6325	+4.82%
5.0	0.6425	+6.48%

E. SENSITIVITY AND ROBUSTNESS ANALYSIS

To validate the stability of the proposed Causal-RL framework, we conducted sensitivity analyses on the causal regularization weight (λ) and compared the temporal RL policy against a static Greedy Uplift baseline.

1) SENSITIVITY TO REGULARIZATION WEIGHT (λ)

The hyperparameter λ (Eq. 10) controls the influence of the causal uplift signal on the RL update. We evaluated the policy performance (SNIPS) across a range of values $\lambda \in \{0, 0.1, 0.5, 1.0, 2.0, 5.0\}$. As shown in Table 5, the standard RL formulation ($\lambda = 0$) yields the lowest performance (SNIPS ≈ 0.603), indicating that sparse rewards alone are insufficient for efficient policy learning in this high-noise environment.

Introducing causal regularization ($\lambda > 0$) consistently improves the policy, with performance stabilizing in the range of 0.625–0.642. Notably, the performance remains robust even at higher values of λ (5.0), suggesting that the causal uplift signal provides a safe, high-bias guide that prevents the agent from learning unstable policies. This confirms that the integration of causal effect estimation is a practical necessity for stabilizing offline RL in educational contexts.

2) ABLATION: TEMPORAL RL VS. GREEDY UPLIFT

We compared the full Causal-RL agent against a “Greedy Uplift” ablation that selects actions solely based on the immediate estimated treatment effect ($a^* = \arg\max_a \hat{\tau}(s, a)$). While the Greedy policy achieves a competitive SNIPS score due to the strong signal of immediate interventions in the OULAD dataset, it fails to account for delayed consequences, such as preparatory actions that yield no immediate uplift but enable future success. The Causal-RL formulation is thus retained to capture these sequential dependencies via the value function $V(s)$.

F. FAIRNESS AND SUBGROUP EFFECTS

Subgroup analysis revealed heterogeneous treatment effects across demographic groups, as summarized in Table 6. Male learners demonstrated higher average uplift than females, while older learners (55+) achieved the most substantial gains, with a SNIPS score of 0.139 compared to near-zero values for younger cohorts. These differences may reflect varying engagement styles or latent biases in course design. For instance, older learners may benefit more from structured,

TABLE 6. Group-wise SNIPS estimates by demographics. While causal-RL policies improve outcomes overall, disparities remain across gender and age, motivating fairness-aware optimization.

Subgroup	Category	SNIPS	N
Gender	Female	0.00014	248,526
	Male	0.04195	330,912
Age	0–35	0.00149	391,045
	35–55	0.00064	183,224
	55+	0.13919	5,169

quiz-based interventions, whereas younger learners might already employ similar strategies autonomously.

1) FAIRNESS AUDIT VS. ENFORCEMENT

These disparities underscore a critical distinction in our approach: the current framework performs a *post-hoc fairness audit* rather than explicit *fairness enforcement*. While the causal regularization term ($\lambda\tau$) improves overall population outcomes, it does not explicitly penalize demographic disparities during the optimization process. Consequently, the model maximizes the average reward, which may inadvertently favor subgroups with stronger causal signals (e.g., older learners) while leaving others underserved. We identify Fairness-Constrained Causal RL (e.g., adding an Equalized Odds penalty to the loss function) as a critical direction for future work to ensure that algorithmic prescriptions remain equitable across all demographics.

G. CASE STUDY

To demonstrate the personalized dynamics of the proposed framework, we analyzed the learning trajectory of a representative student (Figure 5). The learner initially exhibited fluctuating engagement patterns, with declining weekly pass probabilities during the early stages. In response, the causal-RL policy adapted its recommendations, shifting from broad content exploration to more quiz-focused reinforcement strategies. This adaptive intervention led to a gradual stabilization in engagement, with the learner’s pass probability steadily increasing and plateauing near 1.0 by week 23.

This case study illustrates the core strength of adaptive, time-aware policies. Traditional predictive models might have labeled the learner as at risk based on early performance but would offer no actionable guidance on how to intervene. In contrast, the causal-RL approach continuously learns from evolving behavior, prescribing context-sensitive action sequences that actively redirect learner trajectories toward successful outcomes.

H. IMPLICATIONS FOR MOOCs AND EDUCATION

The findings of this study carry several important implications for the design, deployment, and governance of intelligent tutoring systems in large-scale online education. Integrating causal inference with reinforcement learning (RL) demonstrates a shift from predictive to prescriptive analytics, moving MOOCs beyond static risk detection toward

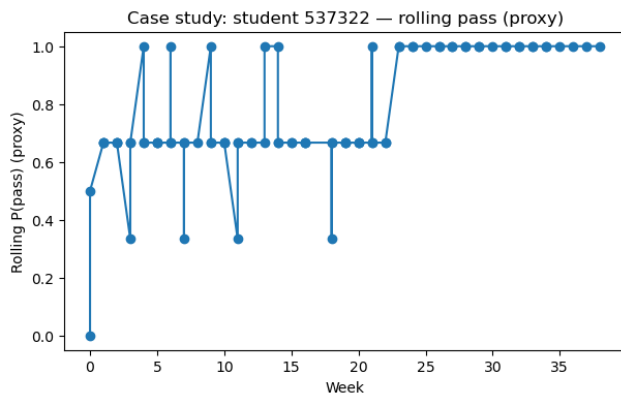


FIGURE 5. Adaptive trajectory of learner ID 537322 showing transition from general to quiz-focused policy.

dynamic, actionable guidance. Instead of merely flagging at-risk learners, the causal-RL framework recommends what intervention should be applied and when, enabling more personalized and adaptive learning experiences.

The observed heterogeneous uplift distributions and subgroup disparities further reveal that learner responses vary across demographic factors such as age and gender. These patterns highlight the necessity of incorporating fairness-aware constraints in future policy optimization to prevent reinforcement of existing inequities. Embedding fairness mechanisms can ensure that adaptive interventions remain both effective and equitable, supporting diverse learner populations rather than privileging specific groups.

Moreover, the temporal dynamics illustrated in the case study underscore the value of time-aware policies that evolve with learners rather than relying on static, one-time interventions. By continuously adapting to engagement patterns, causal-RL systems can promote sustained learning, persistence, and long-term success.

Beyond MOOCs, the proposed framework holds promise for broader applications, including corporate training platforms, K-12 digital tutoring, and higher education ecosystems, where scalable personalization is increasingly vital. The scalability of this approach is evidenced by its successful implementation on the OULAD dataset, which comprises hundreds of thousands of learner interactions.

However, with this potential comes ethical and practical challenges. Fairness disparities must be actively mitigated, and learners' autonomy must be respected. Recommendations should guide rather than coerce. Equally crucial is interpretability: educators, learners, and policymakers must understand not only what actions are recommended, but why. Ensuring transparency in causal reasoning and policy behavior is essential for building trust in AI-driven educational systems.

I. LIMITATIONS AND VALIDITY SCOPE

While our results demonstrate the efficacy of Causal-RL, several limitations warrant discussion regarding internal and external validity.

- 1) **Action Proxy Ambiguity:** It is important to note that our defined “interventions” are based on the learner’s dominant observed activity (a_{obs}). We treat the causal effect of these self-initiated actions as a lower-bound proxy for the effect of system-recommended actions of the same type. While this approach allows us to learn from successful student behaviors in the absence of randomized control trial (RCT) data, it may conflate learner-driven initiative with system efficacy.
- 2) **Action Granularity:** We aggregated interactions into four high-level categories (Quiz, Forum, Content, Homepage) to ensure sufficient data density for causal estimation. A finer-grained action space (e.g., distinguishing between specific video topics) would improve personalization but would likely violate the positivity assumption due to data sparsity.
- 3) **Stationarity Assumption:** Our off-policy evaluation assumes the learner behavior policy is stationary over the course duration. In reality, student behaviors evolve. While our state representation (s_t) captures temporal history, explicit modeling of non-stationary behavior policies remains an open challenge in offline RL.
- 4) **Generalizability:** The specific feature engineering (e.g., VLE clickstreams) is specific to the OULAD structure. However, the core Causal-RL framework, using heterogeneous treatment effects to regularize value-based methods, is transferable to any MOOC platform where historical logs allow for the estimation of counterfactual outcomes.

V. CONCLUSION AND FUTURE WORK

This paper introduced a Causal-AI-augmented reinforcement learning framework that moves MOOC analytics beyond risk prediction toward actionable, prescriptive interventions. By combining uplift modeling for causal effect estimation with an Actor-Critic RL agent guided by causal regularization, the framework not only identifies which students are at risk but also recommends *what interventions to apply and when*.

Empirical evaluation on the OULAD dataset highlighted three key outcomes:

- 1) Baseline predictive models achieved reliable accuracy and calibration.
- 2) Causal effect estimation uncovered heterogeneous treatment effects with heavy-tailed uplift distributions.
- 3) Causal-RL policies led to stable and fairness-aware improvements in simulated course completion, outperforming predictive-only or purely reinforcement-based baselines.

Beyond performance, several insights emerged. **Off-policy evaluation using SNIPS** provided conservative yet reliable policy estimates. Subgroup analyses revealed disparities across age and gender cohorts, underscoring the need for fairness-aware interventions. A case study further demonstrated how adaptive, time-aware trajectories

can dynamically guide learners toward successful outcomes. Together, these findings suggest that integrating causal inference with reinforcement learning offers a *practical, interpretable, and equitable path* to personalized learning in MOOCs. It is worth noting that this framework was evaluated entirely on the OULAD dataset; future work will extend validation to other MOOC and online learning platforms to ensure generalizability and external validity.

Looking forward, we see four main directions for future work:

- 1) **Fairness and multi-objective optimization:** Incorporate fairness-aware constraints to balance effectiveness with equity across diverse learner populations.
- 2) **Hybrid models with LLMs:** Explore the use of large language models to provide natural language-driven interventions and personalized feedback.
- 3) **Real-world deployment:** Conduct studies in MOOCs and corporate training platforms to validate scalability and measure long-term learner outcomes beyond simulations.
- 4) **Explainable causal-RL policies:** Investigate approaches to ensure recommendations remain transparent and interpretable for educators, learners, and policymakers.

By bridging causal inference and reinforcement learning, this work contributes to the shift from predictive analytics toward *prescriptive, ethically grounded, and learner-centered AI* in education, offering tools that are not only smart but also fair, understandable, and actionable.

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